

Autonomous Theory Construction Laboratory

Lecture 4: Attribution and Reproducibility

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Overview

Contributions are recorded by operation and output so that authorship and technical assistance remain distinguishable.

1 Problem definition

The scientific question and the admissible output are specified before any search begins.

$$A_i = (\textit{name}, \textit{build}, \textit{instance}, \textit{role}, [r_0, r_1], \textit{approver})$$

2 Data and model interface

Data schemas and model interfaces are treated as part of the method, not as post-processing details.

$$G_{\textit{contrib}} = (V_{\textit{artifact}}, E_{\textit{transform}})$$

3 Search and evaluation

Proposal and evaluation are kept separate so that a candidate cannot alter its own test.

$$\textit{authororder} = \textit{sort}(\textit{contributionweight}, \textit{firstrevision})$$

4 Reproducibility

Environment, data version, random seed, and decision record are sufficient to reproduce the reported run.

5 Exercise

The exercise asks for a small implementation and an error analysis rather than a polished narrative.

Checks

- Verify dimensions and sign conventions.
- State the boundary conditions used in each limit.
- Report numerical precision separately from model uncertainty.

Reading

- MAS-54 §§1–8
- RLP-12 authorship appendix
- Contribution DAG examples